

Matrices & Matrix Algebra

ANSWER KEY



Matrix addition and scalar multiplication

- 1 Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 2 & 1 \end{bmatrix}$. Find $A + B$ and $2A - B$.

$$A + B = \begin{bmatrix} 1 & 1 \\ 5 & 5 \end{bmatrix}, 2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}, \text{ so } 2A - B = \begin{bmatrix} 2 & 5 \\ 4 & 7 \end{bmatrix}.$$

Matrix multiplication

- 2 Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$. Find AB .

$$AB = \begin{bmatrix} 1(2) + 2(1) & 1(0) + 2(3) \\ 3(2) + 4(1) & 3(0) + 4(3) \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 10 & 12 \end{bmatrix}.$$

Matrix multiplication is not commutative

- 3 For A, B above, compute BA and compare it to AB .

$$BA = \begin{bmatrix} 2(1) + 0(3) & 2(2) + 0(4) \\ 1(1) + 3(3) & 1(2) + 3(4) \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 10 & 14 \end{bmatrix}. \text{ Since } AB = \begin{bmatrix} 4 & 6 \\ 10 & 12 \end{bmatrix} \neq BA, \text{ this confirms matrix multiplication is not commutative in general.}$$

The identity matrix and matrix inverses

- 4 Write the 2×2 and 3×3 identity matrices.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Invertibility of 2x2 matrices

- 5 State the formula for the inverse of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \text{ provided } ad - bc \neq 0.$$

- 6 Find the inverse of $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$.

$$\det A = 3(1) - 1(2) = 1. A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}. \text{ (Check: } AA^{-1} = I_2\text{.)}$$

Inverse of a product

- 7 State the formula for $(AB)^{-1}$ in terms of A^{-1} and B^{-1} .
 $(AB)^{-1} = B^{-1}A^{-1}$ (note the reversed order).

- 8 If $A^{-1} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, find $(AB)^{-1}$.

$$(AB)^{-1} = B^{-1}A^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}.$$

More matrix multiplication practice

- 9 Let $A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}$ (2×3) and $B = \begin{bmatrix} 3 & 1 \\ 2 & 0 \\ 1 & -1 \end{bmatrix}$ (3×2). Find AB .

$$AB = \begin{bmatrix} 1(3) + 0(2) + 2(1) & 1(1) + 0(0) + 2(-1) \\ -1(3) + 3(2) + 1(1) & -1(1) + 3(0) + 1(-1) \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ 4 & -2 \end{bmatrix}.$$

Matrix-vector multiplication

- 10 Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$. Find $A\mathbf{v}$.

$$A\mathbf{v} = \begin{bmatrix} 2(4) + 1(-1) \\ 0(4) + 3(-1) \end{bmatrix} = \begin{bmatrix} 7 \\ -3 \end{bmatrix}.$$

The transpose

- 11 For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ (2×3), write its transpose A^T (3×2).

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \text{ (the columns of } A \text{ become the rows of } A^T\text{).}$$

More 2x2 inverse practice

- 12 Find the inverse of $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$.

$$\det A = 2(3) - 5(1) = 1. A^{-1} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}.$$

Solving a matrix equation

- 13 Use the inverse found above to solve $2x + 5y = 4, x + 3y = 1$ (written as $A\mathbf{x} = \mathbf{b}$).

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 3(4) - 5(1) \\ -1(4) + 2(1) \end{bmatrix} = \begin{bmatrix} 7 \\ -2 \end{bmatrix}. \text{ So } x = 7, y = -2. \text{ Check: } 2(7) + 5(-2) = 4 \checkmark, \\ 7 + 3(-2) = 1 \checkmark.$$

Symmetric matrices

- 14 A matrix is symmetric if $A^T = A$. Is $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$ symmetric?

$$A^T = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} = A, \text{ so yes, } A \text{ is symmetric (its off-diagonal entries mirror each other: } a_{12} = a_{21} = 3).$$

- 15 Is $B = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ symmetric?

$$B^T = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \neq B \text{ (the off-diagonal entries have opposite signs), so } B \text{ is not symmetric.}$$

Powers of a matrix

- 16 Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Find A^2 .

$$A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1(1) + 1(0) & 1(1) + 1(1) \\ 0(1) + 1(0) & 0(1) + 1(1) \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

The trace of a matrix

- 17 The trace of a square matrix is the sum of its diagonal entries. Find $\text{tr}(A)$ for $A = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$.

$$\text{tr}(A) = 2 + 5 = 7.$$

Diagonal matrices

- 18** A diagonal matrix has zeros everywhere except the main diagonal. Write the 3×3 diagonal matrix with entries $2, -1, 5$ on the diagonal.

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

- 19** For the D above, find D^2 .

For a diagonal matrix, squaring just squares each diagonal entry: $D^2 = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 25 \end{bmatrix}.$